HU Collection
ESAT Mock Set D: Mathematics 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	D	B	B	C	C	E	A	F	A

Q	11	12	13	14	15	16	17	18	19	20
Ans	F	D	C	C	A	A	E	F	E	D

Q	21	22	23	24	25	26	27			
Ans	A	B	F	E	D	B	C			

Each solution gives an intended route and names the trap the question was built around. Questions are ordered roughly by increasing difficulty.

Question 1. Answer **B**.

Each transformation acts on x in the current equation. The translation gives $y = \sqrt{x-3}$. A stretch parallel to the x-axis with scale factor $\frac{1}{2}$ replaces x by $2x$, giving $y = \sqrt{2x-3}$. The reflection in the y-axis replaces x by $-x$, giving $g(x) = \sqrt{-2x-3}$, so $g(-5) = \sqrt{7}$.

Applying the stretch as if it acted on the whole bracket $x-3$ gives $\sqrt{-2x-6}$ and the answer 2; translating the wrong way gives $\sqrt{-2x+3}$ and $\sqrt{13}$; replacing x by $\frac{x}{2}$ for the stretch gives $\sqrt{-\frac{x}{2}-3}$, which is undefined at -5.

Question 2. Answer **D**.

The line is $\log_2 y = 3\log_4 x + 2$. Since $\log_4 x = \frac{1}{2}\log_2 x$, this is $\log_2 y = \frac{3}{2}\log_2 x + 2$, so $y = 4x^{3/2}$. Then $y = x$ gives $4x^{1/2} = 1$, so $x = \frac{1}{16}$.

The trap is to treat the horizontal axis as $\log_2 x$, which gives $y = 4x^3$ and the answer $\frac{1}{2}$.

Question 3. Answer **B**.

Here $\dfrac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{8}$ at P. The tangent $y - 4 = \frac{1}{8}(x-16)$ meets $y = 0$ at $x = -16$. The normal has gradient -8, so $y - 4 = -8(x-16)$ meets $y = 0$ at $x = 16.5$. The base TN is 32.5 and the height is 4, so the area is 65.

Using $-\frac{1}{8}$ as the normal gradient (forgetting the reciprocal) puts N at $x = 48$ and gives 128; forgetting the half gives 130.

Question 4. Answer **B**.

$P(1) = -1+9-23+15 = 0$, so $x-1$ is a factor, and division gives $P(x) = -(x-1)(x^2-8x+15) = -(x-1)(x-3)(x-5)$. So $P(x) > 0$ exactly when $(x-1)(x-3)(x-5) < 0$, which is $x < 1$ or $3 < x < 5$. With $x > 0$ the answer is $0 < x < 1$ or $3 < x < 5$.

Forgetting the leading minus sign reverses every region and gives $1 < x < 3$ or $x > 5$; ignoring the small region near the origin gives $3 < x < 5$ only.

Question 5. Answer **C**.

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Write $c = \cos 2x$ (non-zero, as $\tan 2x$ must be defined). Then $\dfrac{\sin^2 2x}{\cos 2x} = \frac{3}{2}$ gives $2(1-c^2) = 3c$, so $2c^2 + 3c - 2 = 0$ and $(2c-1)(c+2) = 0$. Only $c = \frac{1}{2}$ is possible. As x runs over $[0, 2\pi]$, $2x$ runs over $[0, 4\pi]$, giving $2x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$, so $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$, with sum 4π.

Working only over $0 \le 2x \le 2\pi$ gives 2π; summing the values of $2x$ instead of x gives 8π.

Question 6. Answer **C**.

The left side is never negative, so we need $3x > 0$, that is $x > 0$. Then $\left|x^3 - 4x\right| = x\left|x^2 - 4\right|$, and dividing by $x > 0$ gives $\left|x^2 - 4\right| < 3$, so $1 < x^2 < 7$, that is $1 < x < \sqrt{7}$.

At $x = 2$ the left side is $0 < 6$, so 2 is included. Dividing by x without first noting $x > 0$ produces the spurious negative interval.

Question 7. Answer **E**.

A circle with centre $\left(t, \frac{1}{4}t^2\right)$ touching $y = -1$ has radius $\frac{1}{4}t^2 + 1$. The squared distance from the centre to $(0, 1)$ is $t^2 + \left(\frac{1}{4}t^2 - 1\right)^2 = \left(\frac{1}{4}t^2 + 1\right)^2$, so $(0, 1)$ is on every circle: it is the focus of the parabola, whose directrix is $y = -1$.

The circles with $t = 0$ and $t = \pm 2$ already have $(0, 1)$ as their only common point. The point $\left(0, \frac{1}{4}\right)$ is the focus of $y = x^2$, not of this parabola.

Question 8. Answer **A**.

Unfold the path. Reflect B in the x-axis (the last cushion) to $(5, -1)$, then reflect that in the y-axis to $B''(-5, -1)$. The ball's path has the same length as the straight segment AB'', which is $\sqrt{7^2 + 4^2} = \sqrt{65}$. Checking: AB'' meets the y-axis at $\left(0, \frac{13}{7}\right)$ and the next leg meets the x-axis at $\left(\frac{13}{4}, 0\right)$, both on the positive half-axes, so the path is genuine.

Reflecting in only one cushion gives 5 or $\sqrt{53}$.

Question 9. Answer **F**.

Each arc has radius 6 and subtends $\frac{\pi}{3}$ at its centre, so each sector has area $\frac{1}{2} \cdot 36 \cdot \frac{\pi}{3} = 6\pi$, and each circular segment has area $6\pi - 9\sqrt{3}$ (the triangle has area $\frac{\sqrt{3}}{4} \cdot 36 = 9\sqrt{3}$). The region is the triangle plus three segments: $9\sqrt{3} + 3\left(6\pi - 9\sqrt{3}\right) = 18\pi - 18\sqrt{3}$.

Adding three whole sectors to the triangle gives $18\pi + 9\sqrt{3}$, counting the triangle four times; the three segments alone give $18\pi - 27\sqrt{3}$.

Question 10. Answer **A**.

The remainder $r(x) = ax + b$ agrees with $p(x)^3$ at the roots $x = \pm 1$ of $x^2 - 1$. From the given remainder, $p(1) = 4$ and $p(-1) = 2$, so $p(1)^3 = 64$ and $p(-1)^3 = 8$. Then $a + b = 64$ and $-a + b = 8$, so $b = 36$ and $a = 28$.

Equivalently, $(x+3)^3 = x^3 + 9x^2 + 27x + 27$ and replacing x^2 by 1 gives $x + 9 + 27x + 27 = 28x + 36$. Leaving the cube unreduced is not a remainder, since its degree is too large.

Question 11. Answer **F**.

Integrating, $y = x^3 - 3x^2 + c$. The x-axis is a tangent where $y = 0$ and $\dfrac{dy}{dx} = 0$, so at $x = 0$ (giving $c = 0$) or $x = 2$ (giving $c = 4$). There are two possible curves, $y = x^2(x-3)$ and $y = (x+1)(x-2)^2$. For the first, the region is $0 \le x \le 3$ with area $\left|\int_0^3 (x^3 - 3x^2)\,dx\right| = \left|\frac{81}{4} - 27\right| = \frac{27}{4}$. For the second, the region is $-1 \le x \le 2$ with area $\int_{-1}^2 (x^3 - 3x^2 + 4)\,dx = \frac{27}{4}$ as well, so the answer is the same either way. Adding the two cases gives the trap $\frac{27}{2}$; integrating the second curve from 0 to 2 gives 4.

Question 12. Answer **D**.

The moves go east 2, north 1, west $\frac{1}{2}$, south $\frac{1}{4}$, east $\frac{1}{8}$, and so on. The x-coordinate is $2 - \frac{1}{2} + \frac{1}{8} - \cdots$,

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a geometric series with ratio $-\frac{1}{4}$, so it tends to $\dfrac{2}{1+\frac{1}{4}} = \frac{8}{5}$. Likewise $y = 1 - \frac{1}{4} + \frac{1}{16} - \cdots = \frac{4}{5}$. The distance is $\sqrt{\frac{64}{25} + \frac{16}{25}} = \dfrac{\sqrt{80}}{5} = \dfrac{4\sqrt{5}}{5}$.

The total length travelled is 4, which is the trap; $\frac{12}{5}$ adds the coordinates instead of using Pythagoras.

Question 13. Answer **C**.

The polygon splits into n isosceles triangles with two sides 1 and apex angle $\frac{2\pi}{n}$, so its area is $\frac{n}{2}\sin\frac{2\pi}{n}$. The areas increase with n (each polygon gets closer to filling the circle), and for $n = 12$ the area is exactly $6\sin\frac{\pi}{6} = 3$. So the first polygon with area greater than 3 has 13 sides.

No estimate of π is needed; the trap is to stop at 12, where the area equals 3 but does not exceed it.

Question 14. Answer **C**.

Track $L = \log_2(\text{display})$, which starts at 1. Button S doubles L and button H subtracts 1. Work backwards from $L = 100$: an even value may come from half of it (by S) or from one more (by H), and an odd value can only come from one more. The efficient route is

$$1 \to 2 \to 4 \to 8 \to 7 \to 14 \to 13 \to 26 \to 25 \to 50 \to 100,$$

which takes 10 presses.

Fewer is impossible. Six doublings reach at most $2^6 = 64$, so at least seven are needed. A subtraction made when j doublings remain lowers the final value of L by 2^j, so with seven doublings the subtractions must account for $128 - 100 = 28 = 16 + 8 + 4$, which needs at least three of them (three 1s in binary): $7 + 3 = 10$. More doublings only make things worse, since $256 - 100 = 156$ already needs four. An increment button would allow 8 presses via $100 = 1100100_2$, but there is no increment here.

Question 15. Answer **A**.

At $P\left(a, 4 - a^2\right)$ with $0 < a < 2$, the tangent is $y = -2ax + a^2 + 4$. Its intercepts are $a^2 + 4$ on the y-axis and $\dfrac{a^2+4}{2a}$ on the x-axis, so the area is

$$A = \frac{(a^2+4)^2}{4a} = \frac{a^3}{4} + 2a + \frac{4}{a}.$$

Then $\dfrac{dA}{da} = \frac{3}{4}a^2 + 2 - \dfrac{4}{a^2} = 0$ gives $3a^4 + 8a^2 - 16 = 0$, so $(3a^2 - 4)(a^2 + 4) = 0$ and $a^2 = \frac{4}{3}$. This gives $A = \dfrac{(16/3)^2}{8/\sqrt{3}} = \dfrac{32\sqrt{3}}{9}$.

Expanding into powers of a avoids the quotient rule. The tempting $a = 1$ gives $\frac{25}{4}$, which is only slightly bigger.

Question 16. Answer **A**.

The left side $f(x)$ is defined for $-3 \le x \le 5$ and is positive, and

$$f(x)^2 = 8 + 2\sqrt{(x+3)(5-x)} = 8 + 2\sqrt{16 - (x-1)^2}.$$

So f is symmetric about $x = 1$, rises from $f(-3) = 2\sqrt{2}$ to its maximum $f(1) = 4$, then falls back to $f(5) = 2\sqrt{2}$. Each value of c with $2\sqrt{2} \le c < 4$ is taken exactly twice (once on each side of $x = 1$), $c = 4$ once, and other values not at all.

The value $c = 2\sqrt{2}$ is included: it is attained at both endpoints $x = -3$ and $x = 5$.

Question 17. Answer **E**.

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The region is symmetric in both axes, so find the part in the first quadrant and multiply by 4. There, $\sqrt{y} \le 2 - \sqrt{x}$ with $0 \le x \le 4$, that is $y \le (2-\sqrt{x})^2 = 4 - 4x^{1/2} + x$. The area is

$$\int_0^4 \left(4 - 4x^{1/2} + x\right) dx = 16 - \tfrac{8}{3} \cdot 8 + 8 = \tfrac{8}{3},$$

so the total is $\frac{32}{3}$.

The boundary curves inwards (the point $(1,1)$ is on it), so the area is less than that of the square $|x| + |y| \le 4$, which is 32. Forgetting the factor 4 gives $\frac{8}{3}$.

Question 18. Answer **F**.

Let $P = (p, p^2)$ and $Q = (q, q^2)$. The tangent gradients are $2p$ and $2q$, so $4pq = -1$. Then

$$PQ^2 = (p-q)^2 \left(1 + (p+q)^2\right).$$

Writing $s = p + q$, we have $(p-q)^2 = s^2 - 4pq = s^2 + 1$, so $PQ^2 = (1+s^2)^2$ and $PQ = 1 + s^2 \ge 1$, with equality when $p = \frac{1}{2}$, $q = -\frac{1}{2}$.

This is the chord through the focus $\left(0, \frac{1}{4}\right)$ perpendicular to the axis: every such chord passes through the focus.

Question 19. Answer **E**.

Let $g(x) = x^4 - 8x^2 - k$. Since $g'(x) = 4x(x^2 - 4)$, g has a local maximum $g(0) = -k$ and local minima $g(\pm 2) = -16 - k$. Taking the modulus keeps a local maximum of g where $g > 0$, turns a local minimum of g where $g < 0$ into a local maximum, and creates only minima (corners) where g crosses zero.

So $x = 0$ gives a local maximum of $|g|$ exactly when $-k > 0$, and $x = \pm 2$ give two exactly when $-16 - k < 0$. Three maxima need $-16 < k < 0$, giving the 15 integers $-15, \ldots, -1$. At $k = 0$ the point $x = 0$ becomes a minimum of $|g|$ (only 2 maxima), and at $k = -16$ the minima touch the axis (only 1).

Question 20. Answer **D**.

A repeated factor $(x-1)^2$ of $p(x) - 1$ means $p(1) = 1$ and $p'(1) = 0$; similarly $p(-1) = -1$ and $p'(-1) = 0$. So the quadratic $p'(x)$ has roots ± 1: $p'(x) = k(1 - x^2)$ and $p(x) = k\left(x - \frac{x^3}{3}\right) + d$. The conditions $p(1) = 1$, $p(-1) = -1$ give $d = 0$ and $\frac{2k}{3} = 1$, so $p(x) = \frac{1}{2}\left(3x - x^3\right)$ and $p(3) = \frac{1}{2}(9 - 27) = -9$.

A sign slip in k gives $\frac{1}{2}\left(x^3 - 3x\right)$ and the answer 9, but that polynomial has $p(1) = -1$.

Question 21. Answer **A**.

At $A = (a, a^3)$ the tangent $y = a^3 + 3a^2(x - a)$ meets $y = 0$ at $x = a - \frac{a}{3} = \frac{2a}{3}$. So the x-coordinates are $1, \frac{2}{3}, \left(\frac{2}{3}\right)^2, \ldots$, and the triangle at $A = (a, a^3)$ has base $\frac{a}{3}$ and height a^3, with area $\frac{a^4}{6}$.

The areas form a geometric series with first term $\frac{1}{6}$ and ratio $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$, so the total is $\dfrac{1/6}{1 - 16/81} = \dfrac{1}{6} \cdot \dfrac{81}{65} = \dfrac{27}{130}$. As a check, this is less than $\int_0^1 x^3\,dx = \frac{1}{4}$, since the triangles lie under the curve. Using ratio $\frac{2}{3}$ or $\frac{4}{9}$ gives $\frac{1}{2}$ or $\frac{3}{10}$.

Question 22. Answer **B**.

If θ is the angle between the hands, the cosine rule gives the distance d with $d^2 = 34 - 30\cos\theta$, so $2 \le d \le 8$. The minute hand gains 11 full turns on the hour hand in 12 hours, so θ goes through 11 complete cycles.

In each cycle, d takes each value strictly between 2 and 8 twice and the values 2 and 8 once each. The whole numbers $3, 4, 5, 6, 7$ therefore occur 22 times each, and 2 and 8 occur 11 times each (the hands coincide at noon but not at midnight within the period). The total is $5 \times 22 + 2 \times 11 = 132$.

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Counting every whole number 22 times gives 154; leaving out 2 and 8 gives 110.

Question 23. Answer **F**.

The final output at $x = 0$ is b, and it can be read off a difference table. The values of $p(x) = (x(x+1))^3$ at $x = 0, 1, 2, 3, 4, 5$ are $0, 8, 216, 1728, 8000, 27000$. Taking differences five times:

$$8,\ 208,\ 1512,\ 6272,\ 19000; \quad 200,\ 1304,\ 4760,\ 12728;$$

$$1104,\ 3456,\ 7968; \quad 2352,\ 4512; \quad 2160.$$

So $b = 2160$.

Alternatively, the machine lowers the degree by one, sends x^6 to a polynomial that becomes $720x + 1800$ after five uses, and sends $3x^5$ to the constant $3 \times 120 = 360$ after five uses, while $3x^4 + x^3$ disappears; again $b = 1800 + 360 = 2160$. The leading term alone gives 1800; $a = 720$ is the trap of reporting the wrong coefficient.

Question 24. Answer **E**.

Since $f'(x) = 3x^2 - 21$, f decreases on $\left[0, \sqrt{7}\right]$ and increases for $x \geq \sqrt{7}$. If $a > \sqrt{7}$ then f increases on the whole interval and $M(a) = f(a+3) > f\left(\sqrt{7} + 3\right)$, which is larger than values found below. For $0 \leq a \leq \sqrt{7}$ the interval contains $\sqrt{7}$, so the greatest value is at an end: $M(a) = \max\big(f(a), f(a+3)\big)$.

As a increases, $f(a)$ falls and $f(a+3)$ rises, so the larger of the two is least where they are equal. Now $f(a+3) - f(a) = 9a^2 + 27a - 36 = 9(a-1)(a+4)$, so they are equal at $a = 1$, and $M(1) = f(1) = f(4) = -20$.

The trap is the minimum value of f itself, $f\left(\sqrt{7}\right) = -14\sqrt{7}$: that is the least value on the interval, not the greatest. Taking $a = 0$ and reading off $f(3) = -36$ makes the same mistake.

Question 25. Answer **D**.

Doubling a number with d digits gives a number with d or $d+1$ digits, and in the second case the new number lies between 10^d and 2×10^d, so it starts with 1. Conversely, if 2^n lies between 10^d and 2×10^d, then 2^{n-1} lies between $5 \times 10^{d-1}$ and 10^d, so it has one digit fewer. Hence among $2^2, \ldots, 2^{100}$ the powers starting with 1 are exactly those where the digit count increases.

Since $100 \log_{10} 2 = 30.103$, 2^{100} has 31 digits, while 2^1 has 1 digit. The digit count rises 30 times, so the answer is 30. Benford's law suggests about 30%, but the exact count needs this argument; 31 is the digit count of 2^{100}.

Question 26. Answer **B**.

First treat one plate. Light gets through by crossing both faces, possibly after an even number of internal reflections, so the fraction transmitted is

$$T = \left(\tfrac{2}{3}\right)^2 \left(1 + \tfrac{1}{9} + \tfrac{1}{81} + \cdots\right) = \tfrac{4}{9} \cdot \tfrac{9}{8} = \tfrac{1}{2},$$

and since nothing is absorbed, the plate reflects $R = \frac{1}{2}$ overall (directly, $\frac{1}{3} + \frac{4}{9} \cdot \frac{1}{3} \cdot \frac{9}{8} = \frac{1}{2}$).

Now treat each plate as a single unit with $T = R = \frac{1}{2}$. Light passes the first plate, then either passes the second plate or is reflected back and forth between the plates:

$$\tfrac{1}{2} \cdot \tfrac{1}{2} \left(1 + \tfrac{1}{4} + \tfrac{1}{16} + \cdots\right) = \tfrac{1}{4} \cdot \tfrac{4}{3} = \tfrac{1}{3}.$$

Ignoring the repeated reflections gives $\left(\frac{2}{3}\right)^4 = \frac{16}{81}$, or $\frac{1}{4}$ at plate level. Using the single-face value $\frac{1}{3}$ for reflection between the plates gives $\frac{9}{32}$.

Question 27. Answer **C**.

By symmetry, R is made of eight congruent pieces, one in each of the triangles cut off by the axes and the diagonals. In the triangle $0 \leq y \leq x$ the nearest side is $x = 4$, so the condition is

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$x^2 + y^2 < (4 - x)^2$, that is $x < 2 - \frac{1}{8}y^2$. The boundary is the parabola with focus at the centre and directrix $x = 4$.

The parabola meets the diagonal $y = x$ where $x^2 + 8x - 16 = 0$, so at $h = 4\sqrt{2} - 4$. The piece is $0 \le y \le h$, $y \le x < 2 - \frac{1}{8}y^2$, with area

$$\int_0^h \left(2 - y - \tfrac{1}{8}y^2\right) dy = 2h - \tfrac{1}{2}h^2 - \tfrac{1}{24}h^3.$$

With $h^2 = 48 - 32\sqrt{2}$ and $h^3 = 320\sqrt{2} - 448$, this is $\dfrac{32\sqrt{2} - 40}{3} = \dfrac{8\left(4\sqrt{2} - 5\right)}{3}$. Hence the area of R is $\dfrac{64\left(4\sqrt{2} - 5\right)}{3} \approx 14.0$.

Using only the two sides $x = \pm 4$ gives the region between two parabolas, of area $\frac{64}{3}$. Forgetting to multiply by 8 gives $\frac{8\left(4\sqrt{2}-5\right)}{3}$, and multiplying by 4 instead gives $\frac{32\left(4\sqrt{2}-5\right)}{3}$.

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